



Discovery before deployment

A problem- and product-discovery method for social impact organizations building new AI products for the people they serve

Dan Grimm · dangrimm.ai

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WHY NOW

The model was ready. The fit was not.

Two randomized trials, two good AI tools, no change in the outcome that mattered.

REAL CASE

AI math tutor, 18 Tennessee middle schools

ACCESS · MOVED

96%

of students tried the tutor

OUTCOME · ABOUT THE SAME AS WITHOUT AI

0.06–0.08 SD

a year, as with Khan Academy alone. The median student used the tutor on 1 in 3 practice days.¹

REAL CASE

AI clinical copilot, Nairobi primary care

OUTPUT · MOVED

1.74x

the odds of recording an appropriate diagnosis

OUTCOME · DID NOT MOVE

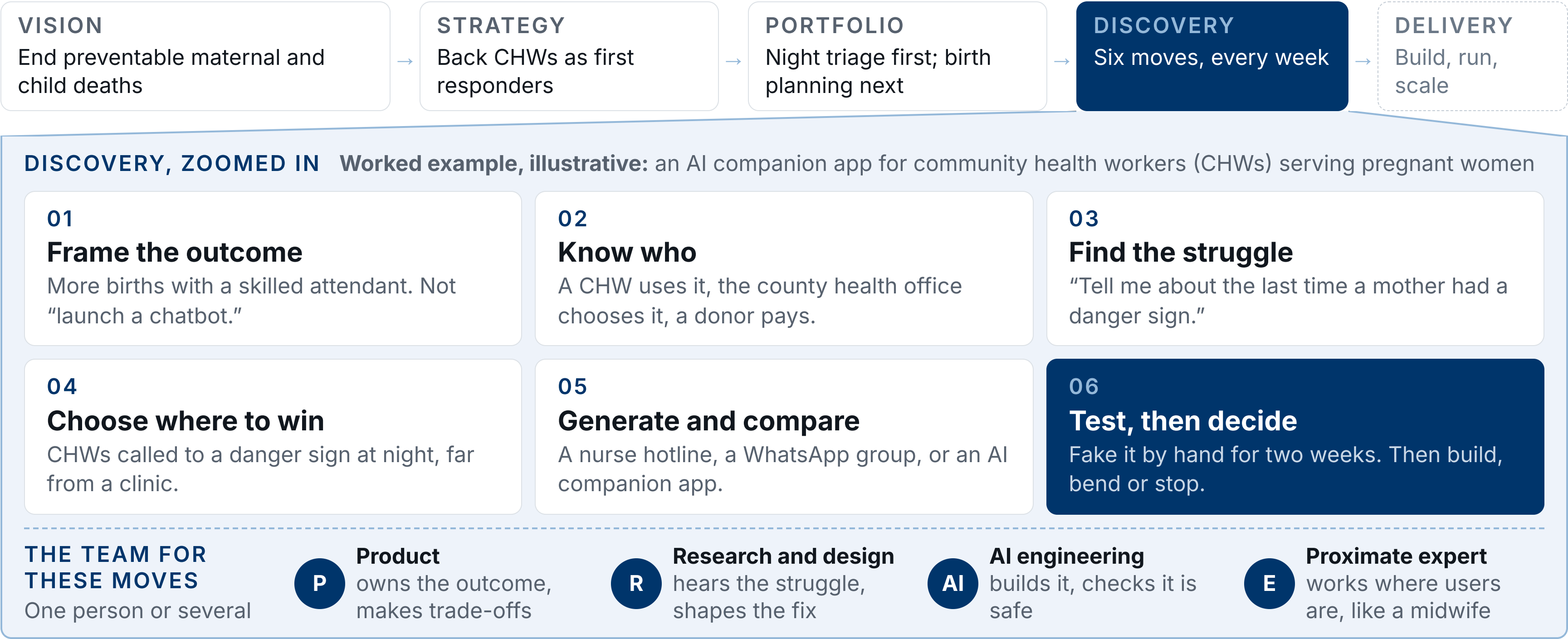
2.2% vs 2.0%

treatment failure at 14 days, with AI vs. without²

Both tools worked. Neither moved the outcome on its own.

¹ Oreopoulos & Low, “One Click Away,” EdWorkingPaper 26-1551 (2026). ² Agweyu et al., *Nature Medicine* (2026).

Six moves, every week, by a team that owns the outcome



The user, the customer and the payer are **rarely the same person**

Worked example, illustrative

User

Touches the product
The CHW, on a low-cost smartphone

Beneficiary

Whose outcome changes
The mother and her newborn

Customer

Decides to adopt it
The county health office

Payer

Funds it
A donor, for year one

REAL CASE

Ask users directly. The payer's dashboard will not tell you this.

When 60 Decibels asked 450 Kenyan farmers about Digital Green's FarmerChat, women recommended it more than men (NPS 75 vs. 57), and 61% used it meaningfully, against a 27% local benchmark.¹

¹ [60 Decibels with Digital Green, FarmerChat farmer survey, Kenya \(n=450\).](#)

Segment by need, not by profile

"I'm sixty-four years old. I'm six feet eight inches tall. My shoe size is sixteen. My wife and I have sent all our children off to college. I live in a suburb of Boston and drive a Honda minivan to work. ... But these characteristics have not yet caused me to go out and buy the *New York Times* today." Clayton Christensen¹

Worked example, illustrative

Who	What they need now	What you would build
Pre-qualifier: community health workers in rural East Africa	Too broad to say	A general health chatbot
CHWs called to a danger sign at night, far from a clinic	Decide fast: refer, call or wait	Triage support with a nurse handoff
CHWs whose mothers think home birth is safer	Help a mother plan a facility birth	A birth-planning script and checklist

Same CHWs, different needs, different products. A profile tells you whom to count, not what to build. More ways to segment: appendix A3.

¹ Christensen, Hall, Dillon & Duncan, *Competing Against Luck* (2016); the same story in [“Know Your Customers’ Jobs to Be Done,” HBR \(2016\)](#).

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People hire a product to make progress in a specific circumstance

	Real case: milkshakes ¹	Worked example, illustrative
Circumstance	A long, boring commute. 40% of shakes sold first thing in the morning, to go.	A call at night: a mother is bleeding, the clinic is two hours away.
The job	Make the drive more interesting, with one free hand.	Get her to the right care in time, and keep the family's trust.
Real competitor	Whatever else fills the commute	Phoning a nurse she knows, the birth attendant, or waiting for morning
So build for	The commute, not milkshake lovers	The whole job: triage plus a way to arrange transport

To find the job, ask about the last switch: push, pull, anxiety and habit (appendix A2).

¹ "Clay Christensen's Milkshake Marketing," HBS Working Knowledge (2011).

Ask about the last time, not about the future

Story-based interviews with CHWs, the users, every week, by the team that builds.¹

Don't ask a CHW

"Would you use an AI app on your home visits?"

You hear what people think you want to hear.

Ask a CHW

"Tell me about the last time a mother you visit had a danger sign. What did you do first?"

You hear what happened, and where it broke.

What CHWs might say (illustrative). Each one is an opportunity, not yet a solution:

No nurse answered at night

The clinic was two hours away

The family had no cash for transport



¹ [Torres, "Story-Based Customer Interviews," Product Talk \(2024\).](#)

Start with one decision, on a channel people already use

REAL CASE India's AI monsoon forecasts, 2025¹

1

decision: when to sow

1

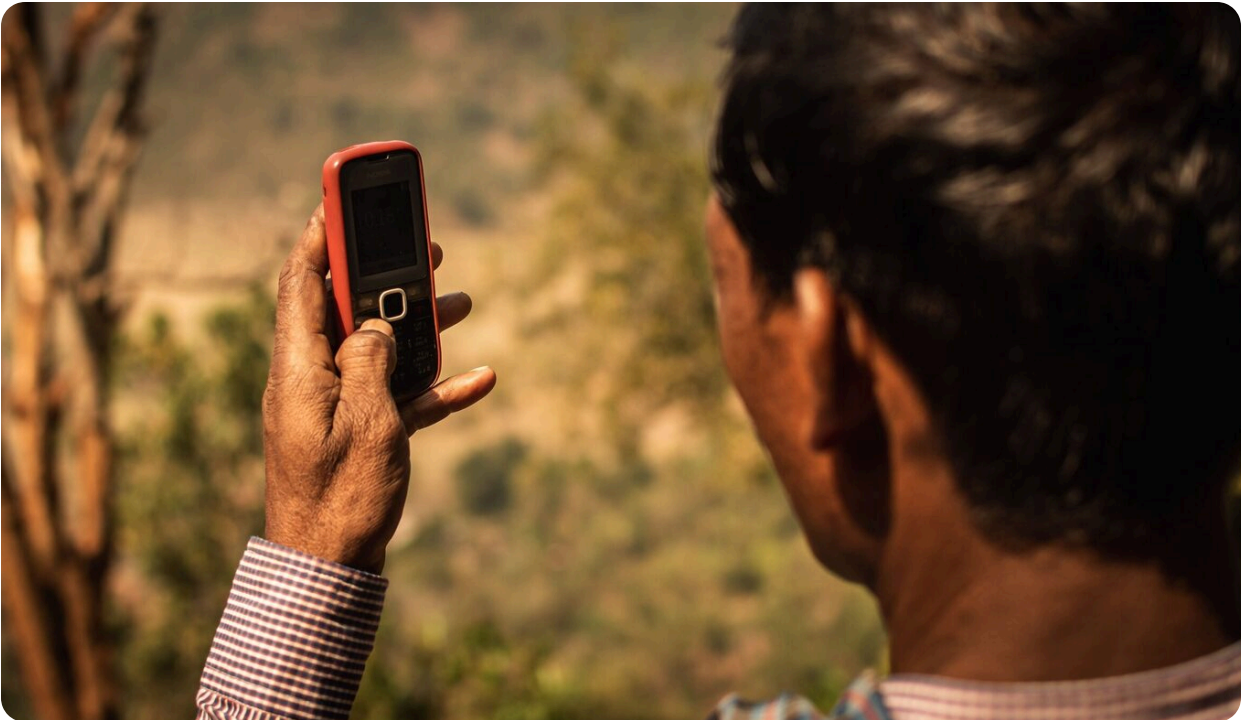
existing channel: the government's farmer SMS service

~38M

farmers reached, in 13 states

31–52%

changed sowing, land prep or inputs, in two surveyed states



Why it worked

One narrow job, a channel farmers already trust, and a gain big enough that going back is unthinkable.

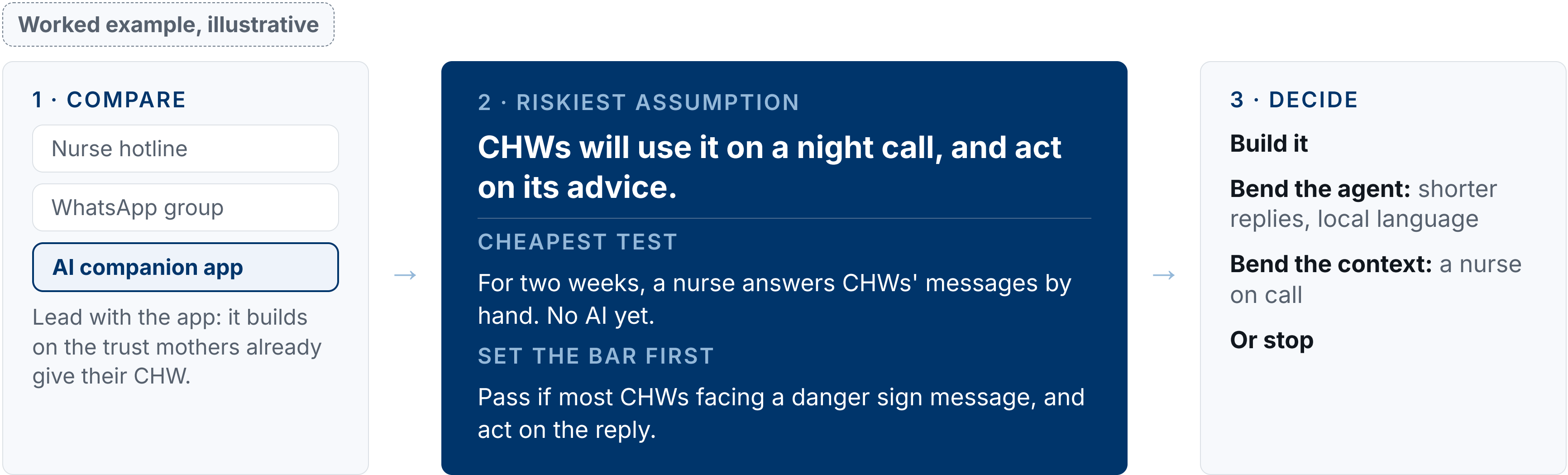
In the worked example (illustrative)

Start with night-time danger-sign triage on the CHW's phone. Birth planning can wait.

¹ [UChicago Climate \(2025\)](#), a project partner; the 31–52% comes from one survey, not a national estimate.

Test the riskiest assumption before you build

Compare a few options. Find the risk that would sink the strongest one (desirability, feasibility, viability or strategic fit), test it cheaply, then decide.



Check the advice is safe before CHWs rely on it. Then watch model, product and user together, and fund the impact study last.¹

¹ Adapted from [Abrol et al., "An AI evaluation framework for the development sector," The Agency Fund via J-PAL \(2025\).](#)

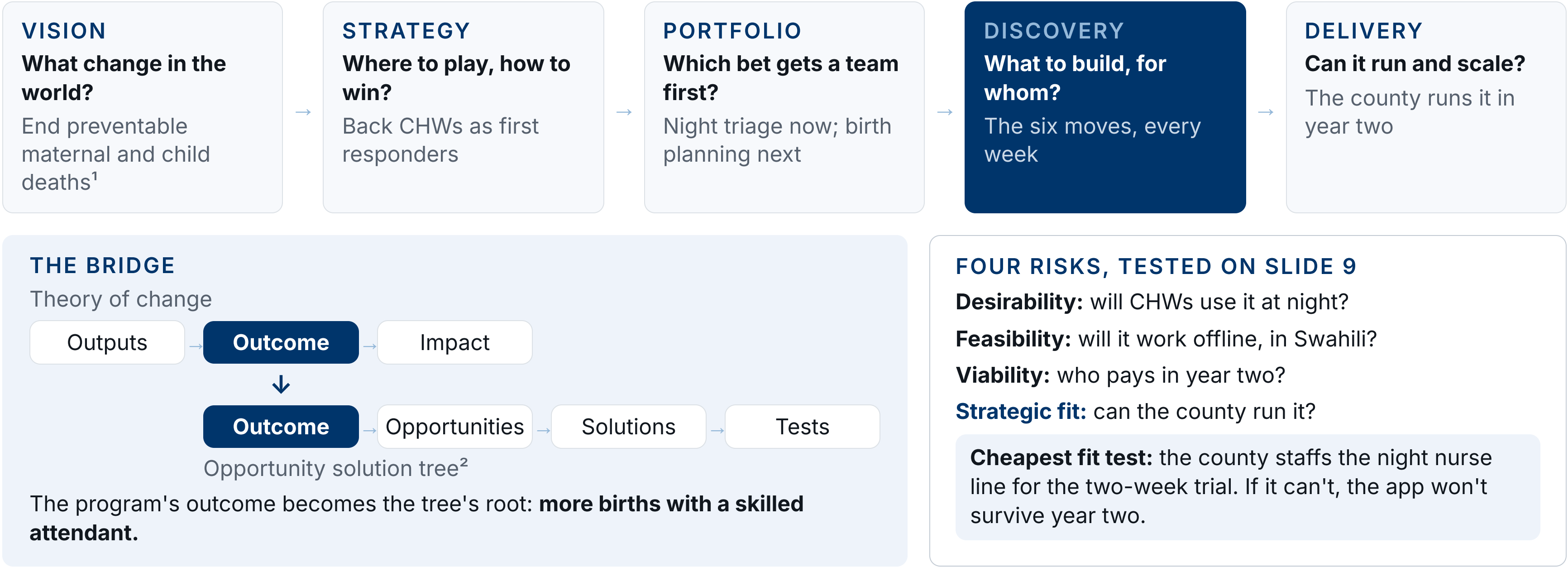
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DISCOVERY BEFORE DEPLOYMENT

Appendix

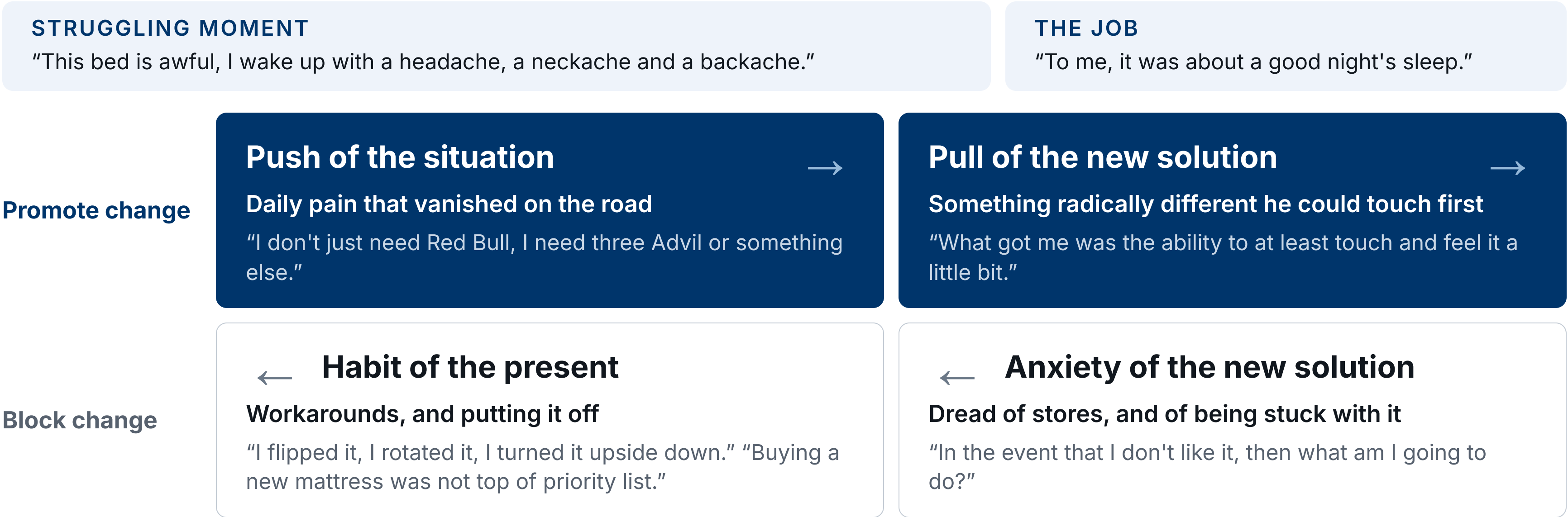
- A1** **Where discovery sits:** vision, strategy and portfolio come first
- A2** **Jobs to be done (JTBD):** four forces decide whether people switch
- A3** **Segmentation:** by need and decision driver, not by demographics

Vision and strategy set the outcome. Discovery finds how to move it.



Four forces decide whether people switch

Real case: the Mattress Interview, a switch interview by Bob Moesta and Chris Spiek.¹



People switch when push and pull beat habit and anxiety. Most teams only add pull.²

¹ Moesta & Spiek, "The Mattress Interview," Jobs-to-be-Done Radio. ² Moesta & Engle, *Demand-Side Sales 101* (2020).

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Segment by need and decision driver, not by demographics

↑	Way to segment, better to worse	Example, maternal health (illustrative)
1	Needs, and the job in its circumstance	A danger sign tonight: decide fast whether to refer
2	Distribution	Reached through the county health office, or an NGO
3	Adoption path	Starts from a supervisor's training, or a peer's tip
4	Decision maker's motivations	County office wants outcomes, or wants reports
5	Role	CHW, CHW supervisor or nurse
6	Sophistication	Has used a health app before, or never
7	Lifecycle	New, active or lapsed users of the app
8	Scale and setting	Catchment of a dispensary, health center or hospital
9	Geography	Rift Valley or coastal counties

PRE-QUALIFY FIRST

Who is in scope at all: community health workers in rural East Africa. A filter, not a segment.

AVOID AS SEGMENTS

Demographics, psychographics, composite personas, early vs. late adopters.

THEN ASK

Can we differentiate with them? Reach them? Will others look like them later?

REAL CASE

Uttar Pradesh, 5,613 women: **69%** of the variance in where they gave birth was individual, **17%** geographic. Segments split on beliefs and planning, not caste.¹